

PKCERT AI & Software Development Internship

Task 07 – Classification Models: Logistic Regression, Decision Trees & Random Forests

Objective:

Develop practical skills in building and evaluating classification models using Logistic Regression, Decision Trees, and Random Forests. Learn to evaluate model performance using classification metrics.

Total Marks: 100

Part A – Logistic Regression (30 Marks)

1. Select a public classification dataset.
2. Build a Logistic Regression model.
3. Explain the working principle and a real-world application of Logistic Regression.

Part B – Classification Metrics (30 Marks)

1. Evaluate the model using Accuracy, Precision, Recall, and F1-Score.
2. Generate and interpret a Confusion Matrix.
3. Explain the importance of each evaluation metric.

Part C – Decision Trees & Random Forests (30 Marks)

1. Implement a Decision Tree classifier.
2. Implement a Random Forest classifier.
3. Compare their performance and discuss advantages and limitations.

Part D – Comparative Analysis (10 Marks)

1. Compare Logistic Regression, Decision Tree, and Random Forest using evaluation metrics.
2. Recommend the most suitable model for your dataset with justification.

Evaluation Criteria

Assessment Area	Marks
Logistic Regression	30
Classification Metrics	30

Decision Tree & Random Forest	30
Comparative Analysis	10